Mutually Exclusive Events and Either/Or

Name _____ Date _____ No calculator Version 1

Scan to practise this topic online

KEY WORDS event *(takahanga)* · mutually exclusive · either/or · overlap · double counting · outcome *(putanga)*

A Can both happen? ●○○

Remember: Two events are mutually exclusive if they cannot both happen at the same time. Rolling a 2 and rolling a 5 on one dice are mutually exclusive.

1. An ordinary dice is rolled once. Are the two events mutually exclusive?

a) 2 and 5

b) Even number and 3

c) Even number and prime number

d) Odd number and even number

e) Number greater than 4 and multiple of 3

f) Number less than 3 and number greater than 4

g) Square number and odd number

h) Multiple of 3 and 4

i) Prime number and 1

j) Number greater than 2 and number less than 5

k) Multiple of 4 and odd number

l) Multiple of 3 and even number

B Adding probabilities ●●○

Remember: If two events are mutually exclusive, add their probabilities to find the probability of one or the other.

2. An ordinary dice is rolled. Find the probability of getting:

a) a 2 or a 5

b) an even number or a 3

c) a number less than 3 or a number greater than 4

d) a multiple of 3 or a 4

e) a prime number or a 1

f) a multiple of 4 or an odd number

3. Cards numbered 1 to 10 are shuffled and one card is picked. Find the probability of getting:

a) a number less than 3 or a number greater than 8

b) a multiple of 5 or a 7

c) a multiple of 4 or a multiple of 3

d) a square number or a 10

e) a prime number or a multiple of 4

f) a multiple of 5 or a multiple of 3

© 2026 Cell Tuition. You may print and photocopy this worksheet for your own classes and children. It may not be changed, sold or posted on other websites.

C When events overlap ●●●

⚠ **Watch out:** If an outcome fits both events, adding counts it twice. Take away the probability of both happening.

4. A spinner has 8 equal sections numbered 1 to 8. Find the probability that it lands on:

a) an even number or a number greater than 5

b) a prime number or an odd number

c) a multiple of 4 or an even number

d) a square number or a number less than 3

5. An ordinary dice is rolled. Find the probability of getting:

a) an even number or a prime number

b) a number greater than 3 or an even number

c) an odd number or a multiple of 3

d) a square number or a number less than 3

6. Cards numbered 1 to 12 are shuffled and one card is picked. Find the probability of getting:

a) an even number or a multiple of 3

b) a prime number or an odd number

c) a multiple of 4 or an even number

d) a number greater than 8 or a multiple of 5

D Problems ●●●

7. In a class of 30 students, 12 play football and 10 play netball. 4 of these students play both. One student is chosen at random. Find the probability that the student plays football or netball.

8. A bag holds 5 red, 3 blue and 4 green counters. One counter is picked. Find the probability that it is red or green.

9. The probability that a bus is early is 0.15, and the probability that it is on time is 0.6. Find the probability that it is early or on time.

★ Extension

10. Cards numbered 1 to 20 are shuffled and one is picked. Find the probability of getting an even number or a multiple of 5. First list the numbers that are both.

© 2026 Cell Tuition. You may print and photocopy this worksheet for your own classes and children. It may not be changed, sold or posted on other websites.

Mutually Exclusive Events and Either/Or

Name _____ Date _____ No calculator Version 2

Scan to practise this topic online

KEY WORDS event *(takahanga)* · mutually exclusive · either/or · overlap · double counting · outcome *(putanga)*

A Can both happen? ●○○

Remember: Two events are mutually exclusive if they cannot both happen at the same time. Rolling a 2 and rolling a 5 on one dice are mutually exclusive.

1. An ordinary dice is rolled once. Are the two events mutually exclusive?

a) 1 and 6

b) Odd number and 4

c) Odd number and prime number

d) Even number and odd number

e) Number less than 3 and even number

f) Number less than 2 and number greater than 5

g) Square number and even number

h) Multiple of 3 and 5

i) Prime number and 4

j) Number greater than 3 and number less than 6

k) Multiple of 4 and prime number

l) Multiple of 3 and odd number

B Adding probabilities ●●○

Remember: If two events are mutually exclusive, add their probabilities to find the probability of one or the other.

2. An ordinary dice is rolled. Find the probability of getting:

a) a 1 or a 6

b) an odd number or a 4

c) a number less than 2 or a number greater than 4

d) a multiple of 3 or a 5

e) a prime number or a 4

f) a multiple of 4 or a prime number

3. Cards numbered 1 to 10 are shuffled and one card is picked. Find the probability of getting:

a) a number less than 4 or a number greater than 9

b) a multiple of 5 or a 3

c) a multiple of 4 or a multiple of 3

d) a square number or a 8

e) a prime number or a multiple of 4

f) a multiple of 5 or a multiple of 3

© 2026 Cell Tuition. You may print and photocopy this worksheet for your own classes and children. It may not be changed, sold or posted on other websites.

C When events overlap ●●●

> ⚠ **Watch out:** If an outcome fits both events, adding counts it twice. Take away the probability of both happening.

4. A spinner has 8 equal sections numbered 1 to 8. Find the probability that it lands on:

a) an odd number or a number greater than 4

b) a prime number or an even number

c) a multiple of 3 or an odd number

d) a square number or a number less than 5

5. An ordinary dice is rolled. Find the probability of getting:

a) an odd number or a prime number

b) a number greater than 2 or an even number

c) an even number or a multiple of 3

d) a square number or a number less than 4

6. Cards numbered 1 to 12 are shuffled and one card is picked. Find the probability of getting:

a) an even number or a multiple of 4

b) a prime number or a number less than 6

c) a multiple of 3 or an odd number

d) a number greater than 7 or a multiple of 4

D Problems ●●●

7. In a class of 40 students, 18 play football and 15 play netball. 5 of these students play both. One student is chosen at random. Find the probability that the student plays football or netball.

8. A bag holds 6 red, 5 blue and 4 green counters. One counter is picked. Find the probability that it is red or green.

9. The probability that a bus is early is 0.15, and the probability that it is on time is 0.6. Find the probability that it is early or on time.

★ Extension

10. Cards numbered 1 to 20 are shuffled and one is picked. Find the probability of getting a multiple of 3 or a multiple of 4. First list the numbers that are both.

© 2026 Cell Tuition. You may print and photocopy this worksheet for your own classes and children. It may not be changed, sold or posted on other websites.